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To cite this article before publication: Josephine Henze *et al* 2026 *Environ. Res. Lett.* in press <https://doi.org/10.1088/1748-9326/aeac70>

Manuscript version: Accepted Manuscript

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Decarbonizing Under Water Stress: Water-Energy-Land Nexus Insights from Chile’s 2050 Carbon Neutrality Pathway

Josephine Henze^{1,*}, Silvia R. Santos da Silva², Felipe Feijoo³, Francisco Flores⁴, Fernando Miralles-Wilhelm², Paelina DeStephano⁵, Page Kyle⁵, and Homero Paltán^{1,6}

¹ School of Geography and the Environment, University of Oxford, Oxford, UK
² University of Maryland Center for Environmental Science, Cambridge, USA
³ School of Industrial Engineering, Pontificia Universidad Católica de Valparaíso, Valparaíso, Chile
⁴ Faculty of Mechanical Engineering and Naval Architecture, University of Zagreb, Zagreb, Croatia
⁵ Pacific Northwest National Laboratory, Richland, USA
⁶ Instituto de Geografía, Universidad San Francisco de Quito, Quito, Ecuador

Received xxxxxx
Accepted for publication xxxxxx
Published xxxxxx

Abstract

Anthropogenic climate change is exacerbating water scarcity in Mediterranean climate regions, such as Central Chile, which already has a low annual precipitation rate, increasing the likelihood of a potentially permanent “mega-drought”. While carbon-neutrality policies often combine energy- and land-sector measures, their specific implications for water-energy-land interactions in Chile remain insufficiently explored. This study presents the first national-level application of GCAM-Chile to assess how combined energy- and land-sector carbon-neutrality pathways may reshape sectoral water-demand patterns in an already water-stressed context. We examine one reference scenario and eight alternative carbon-neutrality pathways combining energy-sector decarbonization with different levels of land-sector mitigation while holding basin water supply static, so results reflect changes in water demand rather than future water availability. Findings indicate that renewable energy-enabled decarbonization substantially reduces electricity-sector water withdrawals by decoupling power generation from thermal cooling demand, offering direct water-demand co-benefits in a water-scarce context. Scenarios that assume the emergence of low-cost carbon capture and storage technologies, achieve smaller reductions in electricity water withdrawals due to the higher water demands associated with fossil generation equipped with these technologies. However, total water withdrawals remain dominated by agricultural irrigation, which continues to grow under all scenarios. Land-sector mitigation reduces irrigation demand by constraining agricultural production, but introduces trade-offs with food production. The implications of these interactions could instigate a new policy landscape of water-energy-land interactions in Chile’s near future, necessitating further attention to nexus implications of climate policies in water-scarce contexts.

Keywords: climate change policies, water-energy-land nexus, carbon neutrality, multi-sector dynamics, water scarcity

1. Introduction

To avert the worst impacts of climate change, the global community needs to reach net-zero carbon dioxide (CO₂)

* Corresponding author. E-mail address: josephinehenze@gmail.com

simultaneously decarbonizing the energy sector, which is the dominant source of CO₂ emissions globally, while enhancing carbon sequestration in the land sector to reach carbon neutrality (CN) [4,5]. However, the cross-sectoral implications of CN policies for adjacent, but interdependent, sectors, such as land and water, are seldom collectively considered.

Nonetheless, existing research clearly demonstrates how water, energy and land (WEL) systems are deeply interconnected with synergistic and competitive impacts [6–8]. The risk of neglecting WEL interdependencies, known in the literature as the WEL nexus, is especially acute in arid and water-scarce regions like the Middle East and North Africa where energy and food production can be constrained by water availability [9]. CN policies often rest optimistically on the assumption that there is sufficient water availability which may create unintended consequences for the water sector [10]. Consequently, arid countries may need to be more attuned to WEL interactions and factor in how CN policies add a new level of complexity. While the WEL nexus encompasses multiple bidirectional interactions, this study adopts a water-centric entry point, focusing on how CN policies propagate through energy and land systems to shape water demand outcomes.

Chile is a critical case study due to a) its arid climate in the Central and Northern regions, b) an ongoing mega-drought (MD) and c) legally binding 2050 CN target. Since 2010, Chile has experienced the most severe dry period on record, with annual rainfall deficits between 25% and 45% and no significant recovery projected [11,12]. The MD has already constrained hydroelectric energy and agricultural production, causing social and environmental challenges [13]. These pressures are particularly important for agriculture, given the second-largest economic sector in Chile, water scarcity exacerbates existing agricultural constraints as only 1.7% of Chile's land is classified as arable [14]. Against this backdrop of acute supply-side scarcity, mitigation-driven changes in sectoral water demand could further intensify existing pressures. Although climate-driven changes in water supply have been widely studied, demand-side policy interactions remain less well understood, despite evidence that socioeconomic development and changing sectoral demand are major determinants of future water stress [15–18].

Chile aims to reach CN by 2050, defined as a maximum of 65 MtCO₂e in residual GHG emissions to be balanced by sequestration from natural carbon sinks (e.g., forests, reforestation, and afforestation) [19,20]. Key milestones include a maximum peak emissions year in 2025 followed by a complete phase-out of coal-fired power plants by 2040 (Supplementary Note 3) [19]. This target must address a national emissions profile in which energy production accounts for 77% of national GHG emissions, with CO₂ comprising 78% of total GHG emissions in 2018 [19].

However, recent disturbances such as high wildfire prevalence in 2017 have reduced the sequestration potential of Chile's land sector [21]. This could be further stressed by the impacts of climate change and increasing GHG emissions from land use, land use change and forestry (LULUCF) [22–24]. Consequently, many factors add uncertainty in the land sector's effective contribution to Chile's CN goal.

Additionally, new carbon capture and storage (CCS) technologies that address emissions from fossil fuel dependency could intensify water demand in Chile due to energy-intensive capture processes [25]. One global study found an increase in water consumption by 14% and water withdrawal by 17% by 2100 under the imposition of a carbon price and an increasing demand for electricity under CCS policies [26]. Another finds that, for the electricity sector alone, withdrawals may remain stable or decline over the same period due to retirement of once-through cooling systems, even as consumption rises [27]. Chile's economy already relies on water-intensive sectors, such as agriculture, mining and industry, and policies to reach their CN target may further increase water demand [28].

To capture these cross-sectoral nexus interactions, integrated assessment models (IAMs) dynamically assess multiple systems rather than in siloed models [26]. The Global Change Analysis Model (GCAM) is a widely used IAM to explore human and earth-system dynamics and has made significant contributions to climate science, such as the development of the Representative Concentration Pathway 4.5 and the Shared Socioeconomic Pathway 4 storyline [26,29,30]. From the Middle East to Latin America, GCAM has facilitated understanding energy transition pathways, and the impacts on the WEL nexus [9,31]. Santos Da Silva *et al.* explored the implications of the Paris Agreement on the WEL nexus in Latin America using GCAM [32]. Other studies utilized GCAM to evaluate strategies for Colombia to achieve their NDCs [33].

Yet, despite existing efforts, no previous study has assessed how well-intentioned CN policies may influence WEL interactions in Chile. This study provides the first national-level analysis of how CN pathways in Chile propagate through energy and land systems to shape water demand within the WEL nexus. We use a nexus approach to explore how contrasting CN pathways differ in water intensity and how land mitigation shapes agricultural land use and irrigation demand. These combined shifts inform how feasible CN is in a water-stressed context like Chile and what unintended impacts should be considered. Simulating one reference and eight alternative energy and land-based CN pathways using GCAM, we examine the interactions between CN policies and the WEL nexus at the national scale.

Toward this objective, this work employs a special version of GCAM (GCAM-Chile v6; [34]) to address three questions: What WEL interactions affecting water demand emerge in Chile under a no-policy reference pathway until 2050? How do contrasting CN pathways focused on the energy sector

(renewables-dominated vs. CCS-enabled) affect water demand and related WEL interactions in Chile? How do different levels of additional land-sector mitigation alter agricultural land use and irrigation water demand under these energy-sector pathways? In this study, we focus analytically on water demand, as water availability is a central constraint across energy and land systems in Chile. Because basin supply is held static (Section 2.2), we assess how mitigation policy reshapes water demand and its cross-sectoral consequences.

2. Methodological Framework

2.1 GCAM-Chile

GCAM models future global and regional change of five main systems in 5-year intervals: water, energy, land, socio-economics and climate [26]. These systems are represented at different spatial scales, including 32 energy-economy regions, 235 water basins modelled to balance water requirements within each basin, and 384 land regions, which are defined by the intersection of the energy-economy regions with water basins (further details are in Supplementary Note 1) [26].

Sub-regional and regional downscaled versions of GCAM were developed, such as GCAM-USA (downscaled to state level) and GCAM-LA (Latin America) which was further downscaled to a South America Southern (country-level) and Northern group [31,35]. Arriet *et al.* separated Chile as an independent energy-economy region from the broader South American Southern aggregate, creating GCAM-Chile [36].

This study builds on previous research and GCAM-Chile development conducted by Arriet *et al.* [36], Kintner-Meyer *et al.* [37], Flores *et al.* [38], and Feijoo *et al.* [39]. Specifically, we rely on GCAM-Chile v6.0, as updated by Flores *et al.* [38], which incorporates earlier research and model refinements (detailed in Section 2.2).

While these works focused predominantly on energy sector decarbonization strategies, none have explored the compounding impact of land-mitigation strategies, particularly on the water sector. We introduce a novel dimension by explicitly coupling energy decarbonization pathways with varying levels of land mitigation to explore water implications (detailed in Section 2.3).

2.2 Model Calibration and Validation

GCAM-Chile v6.0 has undergone extensive calibration and refinement. Arriet *et al.* [36] adjusted the energy sector with official national energy balance data from the Ministry of Energy for 2015 and 2019 to reproduce observed electricity generation. They also updated the renewable, transport, H2 strategy and electricity modules to reflect current Chilean policies, including renewable cost and potential assumptions, prioritization of electric transport, fuel price penalties, and the exclusion of nuclear power [36].

Further calibrations were conducted by Feijoo *et al.* [39], including expanding details in the industrial sector, calibrating the energy module with IEA energy balances and integrating technology cost data from Chile's Long Term Energy Plan (PELP) [38,39]. Socioeconomic data, such as population and gross domestic product (GDP), were calibrated based on data gathered by the Chilean Ministry of Economy. In the rest of the world, socioeconomic assumptions are based on Shared Socioeconomic Pathway 2 (SSP2) with moderate development and gradual societal changes [40].

Water withdrawals are represented endogenously through sector-specific demand formulations and are standard GCAM defaults [26]. Irrigation demand is determined by crop-level irrigated area and water requirements, while energy-sector water use is linked to technology-specific water intensities for electricity generation and fuel production. Beyond these calibrations, the Chile-specific updates include: (i) the explicit disaggregation of Chile as an independent energy-economy region, with its water basins and land units reassigned from the South America Southern aggregate [36]; (ii) the calibration of electricity generation to observed 2015 and 2019 national energy balances, with updated renewable cost and potential assumptions, prioritization of electric transport, fuel price penalties, and the exclusion of nuclear power [36]; (iii) expanded industrial-sector detail, energy-module calibration to IEA balances, and PELP technology cost data [38,39].

All remaining inputs, including the hydrological supply and land-system parameters for Chile and all assumptions for the rest of the world, retain the values of the GCAM v6.0 core [34]. Consequently, differences across scenarios are attributable solely to the Chilean scenario levers, summarized in *Table 1*. For further details about these assumptions, the reader is referred to Calvin *et al.* [26] and GCAM documentation [34]. The 2015 base year aligns with the final calibration year of the GCAM v6.0 core release, in which historical energy, land, and socioeconomic data are harmonized across all regions, and it cannot be shifted for an individual region without recalibrating the full global model. Developments in Chile after 2015, which falls within the model projection period, are instead captured through the country-specific updates described above. Additionally, the land module allocates land among competing uses according to relative profitability and does not represent episodic environmental shocks [34].

In addition to extensive validation carried out by previous studies, GCAM-Chile outputs under the historical period were checked for coherency with open-source FAO data (AQ-UASTAT and FAOSTAT) (Supplementary Note 2). Finally, water supplies and demands are balanced endogenously at the basin scale through basin-level water prices (Section 2.1). The basin supply curves, however, are derived from historical run-off and remain static over the projection period (Supplementary Note 1). Consequently, the sustained drying trend observed in central Chile is not reflected in future water

availability. The analysis is therefore confined to the demand side of the nexus, evaluating how mitigation policy reshapes cross-sectoral water demand. Since basin water supply is held static, the modelled reductions under RE and LM should be read as changes in demand pressure and not as an accounting of future water availability.

2.3 Scenario Design

This study adopts a scenario-based approach, with results intended as comparative and exploratory rather than predictive. To assess compounding energy and land policies we developed a matrix of one reference and eight alternative scenarios (see *Table 1*). The Reference (REF) scenario assumes no climate mitigation actions by 2050. This scenario is not a “most-likely” scenario but assumes SSP2 trends globally serving as a comparison baseline.

In Kintner-Meyer et al., decarbonization pathways were explored across five countries, including Chile, providing insightful preliminary findings which helped steer national energy policies [37]. Two of the scenarios developed are adopted unchanged in our research and serve as the foundation for our additional scenario development. Namely, a renewables-dominated scenario (RE), where CCS is shut-off, and a CCS-enabled scenario (CCS), where low-cost CCS is allowed to be deployed, to two differing technology pathways to decarbonize Chile’s energy sector [37]. Both scenarios implement the electromobility and hydrogen strategies that are central to Chile’s PELP and NDC (Supplementary Note 3).

Novel to this study, varying levels of land-sector mitigation (LM) are introduced to explore how reliance on terrestrial carbon sinks shifts trade-offs with agriculture and water demand. LM is initially excluded from the RE and CCS pathways to isolate energy-sector effects. Six LM scenarios are constructed on top of the original net-zero scenarios (RE and CCS) to examine the effect of simultaneous land-sector mitigation. In GCAM, land mitigation is implemented as a carbon price on terrestrial carbon set as a share of the fossil carbon price [41,42]. We follow the approach by Binsted et al. [41],

Calvin et al. [42], and Zhao et al. [43]. In addition to pricing fossil fuel and industrial CO₂ emissions (as in the RE and CCS scenarios), the LM scenarios assign an exogenous fraction of the energy and industrial CO₂ price to terrestrial carbon, ranging from 10% (limited, or LLM), to 50% (moderate, or MLM), and 100% (high, or HLM) (Table 1). These studies provide the implementation approach, but not the levels themselves: because this parameter is a relative price, we anchor each level to the relative value that Chilean policy assigns to land carbon.

The limited and high levels are set by two reference points in Chilean policy, and the moderate level lies between them. At the low end, land carbon is currently valued at roughly 7–15% of the official carbon value: land-sector reductions have been paid about US\$5 per tCO₂e under the World Bank agreement [44], while the Social Price of Carbon rose from US\$32.5 to US\$71.1 per tCO₂e between 2020 and 2025 [45], and the green tax does not cover the land sector [46]. We take 10% as the limited level. Slow progress on the 2020 NDC forest commitments (23% and 6.3% of the management and afforestation areas) is consistent with this weak incentive [47]. At the high end, the Framework Law on Climate Change defines neutrality as the balance between emissions and removals and lets certified removals count on equal terms with emission reductions [48]. This implies full parity between land and fossil carbon, equivalent to a universal carbon price [49], which we represent as 100%. The moderate level (50%) is the midpoint.

The scenarios are not intended as predictions or policy prescriptions; here they function as exploratory pathways that isolate how increasing reliance on terrestrial carbon mitigation propagates through land allocation, agricultural production and irrigation demand. Figure S9 highlights the contribution of each policy scenario to cumulative reductions in land-use CO₂ emissions in Chile by 2050. We assess temporal changes over the period 2015–2050, using 2015 (last calibration year in GCAM) as the baseline and 2050 (Chile’s carbon neutrality target) as the end-point comparison year, unless otherwise stated.

Table 1: Description of Scenarios

Overview of the nine modelled scenarios for Chile’s 2050 carbon-neutrality pathways, including the Reference (REF) and eight mitigation scenarios combining energy-sector strategies (Renewables [RE] or Carbon Capture and Storage [CCS]) with varying levels of land-sector mitigation (Limited [LLM], Moderate [MLM], High [HLM]). X = included in scenario design.

Scenario Names	Reference	Renewable Energy Focus	Carbon Capture & Storage Focus	Limited Land Mitigation		Moderate Land Mitigation		High Land Mitigation	
Acronyms	REF	RE	CCS	RE + LLM	CCS + LLM	RE + MLM	CCS + MLM	RE + HLM	CCS + HLM
Scenario Differences									
Emissions									
Coal Phase Out Year 2040	-	X	X	X	X	X	X	X	X

Residual GHG Constraint at 65 MtCO ₂ e by 2050	-	X	X	X	X	X	X	X	X
Renewable Energy									
Low-cost wind and solar	-	X	-	X	-	X	-	X	-
Carbon Capture & Storage									
Economy-wide restriction on CCS	-	X	-	X	-	X	-	X	-
Low-cost CCS	-	-	X	-	X	-	X	-	X
National Energy Strategies									
Electromobility Strategy	-	X	X	X	X	X	X	X	X
Hydrogen Strategy	-	X	X	X	X	X	X	X	X
Land Mitigation									
Land sector mitigation contribution (%)	-	-	-	10%	10%	50%	50%	100%	100%
Common Assumptions									
Updated capacity factors of renewable technologies based on PELP estimations	X								
Population and GDP based on PELP	X								
Hydropower based on PELP Carbon-Neutral Scenario	X								

3. Results

3.1 Electricity generation growth and simultaneous decarbonization of the energy sector

In REF, electricity generation triples between 2015 and 2050 (Figure 1a). In the near term, power generation is primarily met by fossil fuel sources, particularly coal and natural gas. Renewable energy generation expands over time, representing more than half of the power supply by 2050, driven by a reduction in the levelized cost of renewable electricity. This moderates the historically rapid growth in fossil fuel and industry CO₂ emissions, which by 2050 increases 8.5% relative to 2015 (compared to 161% in 2015 relative to 1990; see Figure S5 on REF GHG emissions). Figure S6 shows total GHG and fossil emissions in Chile for all scenarios from 2015 to 2050. Figure S7 also shows primary energy consumption by source for all scenarios.

Compared to the REF scenario, total electricity generation expands marginally in the RE and CCS scenarios as energy end-use sectors are increasingly electrified and reduce consumption of fossil fuels (Figure 1b,c). These scenarios highlight that CN policies demand a major shift in Chile's power sector away from coal towards low-carbon sources, primarily intermittent renewables such as wind and solar. Nonetheless,

oil and gas remain present in the modelled electricity mix, reflecting the outcome of GCAM's technology competition framework under the imposed policy assumptions. While GCAM does not explicitly represent short-term power-system balancing, this outcome is consistent with a continued role for these technologies in meeting residual energy demand as variable renewable generation expands and in supplying difficult-to-decarbonize sectors

As seen in Figure 1c, assuming low-cost CCS technologies become available (Table 1), more fossil fuel use is observed in the power sector when combined with CCS. Moreover, the absence of low-cost renewables (Table 1), relative to the RE scenario, leads to overall lower electricity generation (230 TWh vs. 260 TWh) along with a reduction in solar generation and an increase in fossil generation. Although CCS provides additional flexibility in achieving CN, its future large-scale deployment and low cost remain uncertain due to currently high capital costs, infrastructure requirements for transport and storage, and unresolved questions regarding long-term geological storage capacity in Chile. Note that the LM scenarios are not shown here because they follow similar energy-technology pathways as their counterpart RE and CCS scenarios (Figure S8). Figure S11 shows electricity generation by cooling technology for all scenarios.

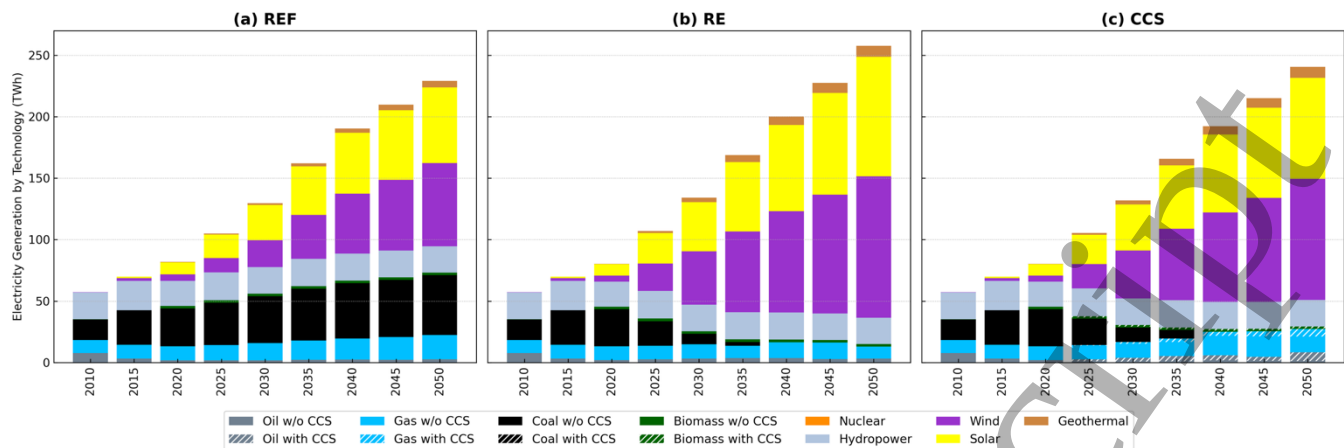


Figure 1. Electricity generation by technology (TWh) for (a) the Reference scenario (REF, no climate policy), (b) the Renewables-dominated scenario (RE), and (c) the Carbon Capture and Storage scenario (CCS) for Chile.

3.2 Land system impacts of energy and land mitigation

In REF, land allocation is dominated by mountainous and desert areas, followed by grassland and pasture, and then forest (Figure 2a). Arable land and forest increase at the expense of grassland and pasture due to the growing societal demand for agricultural and forest products.

Land-use changes remain modest overall in RE and CCS compared to REF, indicating that energy-sector decarbonization alone does not strongly reshape land systems (Figure 2b,c). One key contributing factor is the minimal reliance on bioenergy to decarbonize the power sector as shown in Figure 1. According to prior analyses, large dependence on bioenergy for energy-sector mitigation greatly affects the land and water systems [42,50]. Similarly, forest area remains virtually unchanged, suggesting limited reforestation/afforestation with the absence of explicit LM (Figure 2b,c).

Adding LM drives meaningful land reallocation, with modest expansion of natural ecosystems (grasslands and shrublands), suggesting that natural vegetation could function as a pathway to increase carbon stocks (Figure 2d,e,g,h). Reductions in pasture and arable land, and to a lesser extent forest, highlight competition between carbon sequestration and agricultural production (Figure 2d,e,g,h). This underscores the need for careful consideration of the ambition level of LM

policies given the growing pressures of land and water for agricultural production. Beyond their carbon sequestration potential, the expansion of natural ecosystems such as grasslands and shrublands may provide additional ecosystem services, including biodiversity conservation, soil stabilization, and improved watershed regulation [51,52]. Nonetheless, in the Chilean context, increased shrubland coverage may also elevate wildfire risk under a warming and drying climate in the absence of effective land-sector management, underscoring that LM policies can generate both co-benefits and new environmental challenges [21,53,54].

Patterns are broadly similar between RE+LM and CCS+LM scenarios, with CCS+LM tending to show slightly smaller magnitudes of change in the high and moderate cases. This suggests that availability of power-sector CCS technologies might reduce the level of mitigation effort required by the land sector to achieve the CN goal (Figure S9). However, CCS technologies remain at an early stage of large-scale commercial deployment increasing uncertainty in climate policy decisions.

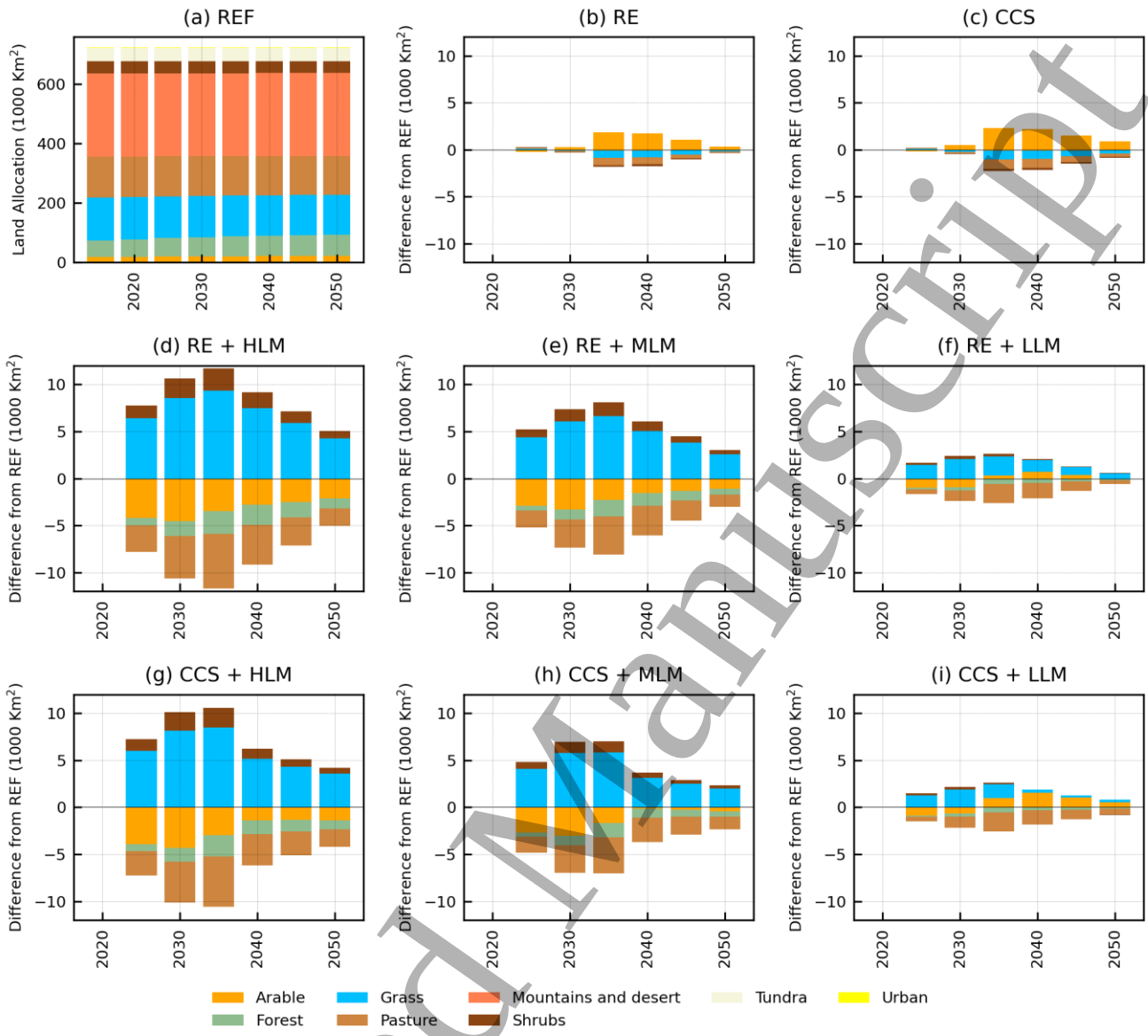


Figure 2. Land allocation (thousand km²) for the Reference scenario (REF, panel a) and change in land allocation relative to this REF baseline for all carbon-neutrality pathways (panels 2b–i) in Chile. Positive values indicate expansion and negative values indicate contraction of each land-use type relative to REF. The changes shown in panels 2b–i are modest compared to 2a.

Relative to the REF scenario, high- and moderate-LM measures produce only minor reductions in forest cover but substantially larger decreases in arable land (Figure 2d,e,g,h), which explains the pronounced decline in agricultural production shown in Figure 3. Note that the two land-use types greatly differ in scale, as total arable land area is considerably smaller than total forest land area (Figure 2a). Since GCAM prices terrestrial carbon based on total carbon stored in vegetation and soils rather than on the resulting emissions from land-use change, land types with high soil carbon stocks respond differently to land mitigation incentives. As a result, grasslands and shrublands in Chile can be favored under

higher terrestrial carbon prices due to their combined vegetation and soil carbon stocks, occasionally leading to small reductions in forest area within the model [55–57]. Consequently, LM policies create stronger shifts in LULUCF. We note that these results should be interpreted within the GCAM modeling framework, where land allocation responds to changes in economic incentives. Apart from responding to the relative profitability of competing uses, future land use choices in the model are anchored to the base-year land shares, which implicitly reflect the ecological characteristics and suitability of a location for a given land type. Consequently, although terrestrial carbon pricing increases the relative

profitability of forests, the largest absolute expansion occurs in grasslands and shrublands because these land-cover types occupy larger base-year shares and are therefore more widely available for expansion within the model, while still contributing to the climate-policy objective. However, these outcomes should not be interpreted as suggesting that reforestation is not a viable climate mitigation strategy in Chile. Rather, our results highlight that, in a country such as Chile, where arable land is limited, competition among land uses may pose important challenges for the implementation of terrestrial climate policies, underscoring the need for integrated land-use planning and management.

Agricultural production in GCAM depends on available arable land and crop yields. Total harvested mass across crop

types is used as a proxy indicator of agricultural output, noting that aggregation across crops with varying yields may skew the metric toward higher-yielding crops. In REF, agricultural production steadily increases across the time horizon driven by socioeconomic demand (Figure 3). The RE and CCS pathways fluctuate around the REF pathway with CCS achieving the largest agricultural outputs in 2050 among the core pathways (Figure 3). Under HLM and MLM, agricultural production initially declines because terrestrial carbon pricing reduces the land available for crop production. After 2030, continued socioeconomic demand partially offsets this initial decline, producing the partial recovery shown in Figure 3, although production remains constrained by the reduced availability of arable land.

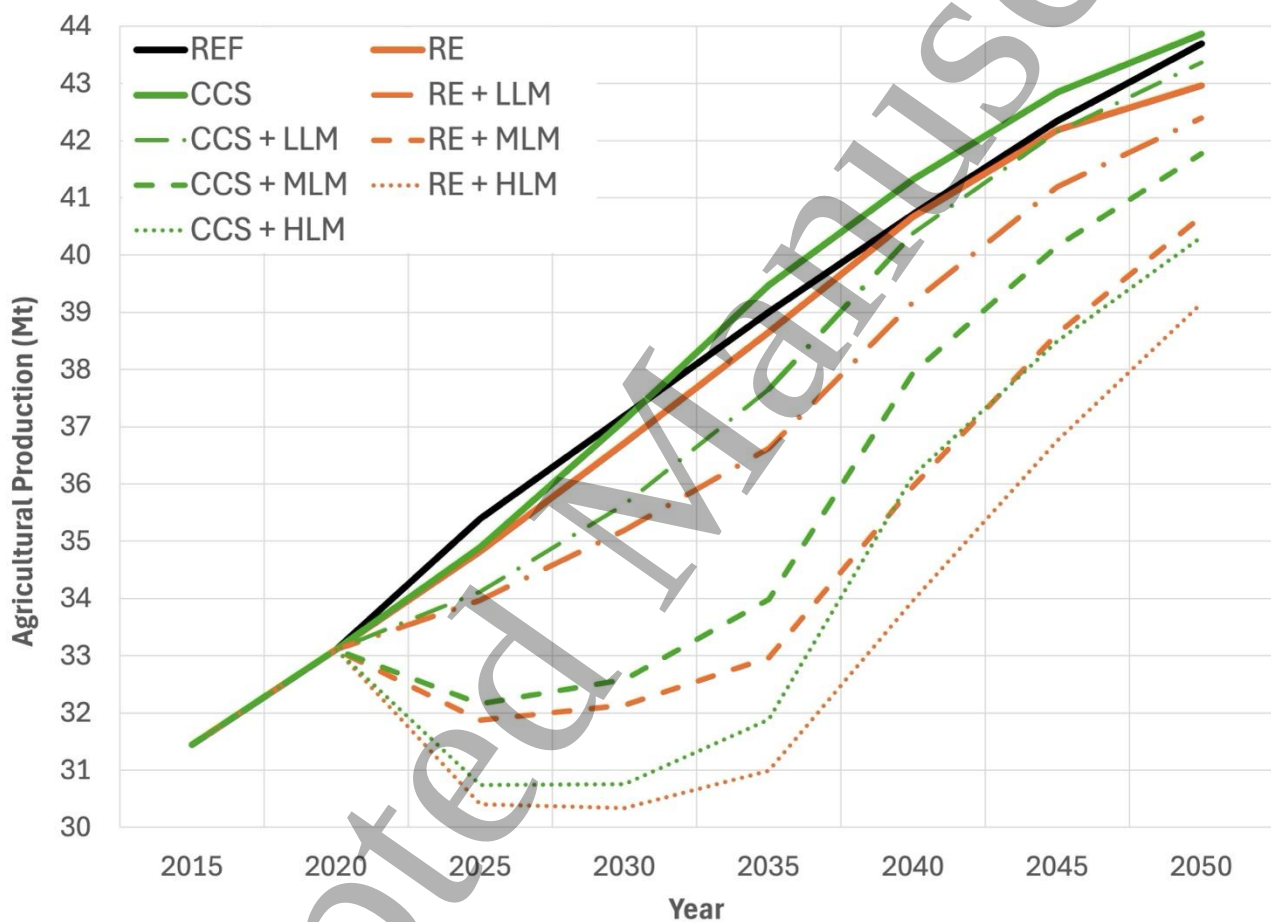


Figure 3. National agricultural production (Mt) across all scenarios. Values represent total annual agricultural production aggregated at the national level for Chile. Non-Land Mitigation scenarios (Reference [REF], Renewables [RE], Carbon Capture and Storage [CCS]) show uninterrupted growth, while Moderate and High Land Mitigation scenarios exhibit temporary declines followed by partial recovery, reflecting land reallocation under terrestrial carbon pricing.

3.3 Shifts in water demand

Agricultural irrigation is the historical dominant withdrawal sector in Chile [47], and together with industrial, municipal and electricity sectors account for approximately 97% of total withdrawals across the modelled period (Figure 4a)

(Figure S10 for all scenarios). In REF, total water withdrawals increase by 36% between 2015 and 2050, primarily in response to population and GDP growth (Figure 4b). As discussed below, the electricity sector is the only sector projected to see a decline in water demand.

Compared to REF, both RE and CCS scenarios reduce total water withdrawals across the projection period, with RE consistently achieving lower withdrawals than CCS (Figure 4b). The changes shown in Figure 4b across all CN scenarios are driven primarily by the irrigation sector and, to a lesser extent, the electricity sector. These drivers are discussed below. Irrigation withdrawals increase over time in both RE and CCS but remain lower than REF in 2050 (Figure 5a).

Scenarios with LM exhibit lower total water withdrawals than non-LM pathways (Figure 4b). By 2050, total water withdrawals decline by ~4–10% relative to REF, with the largest

reductions occurring under HLM scenarios (~8–10%) (Figure 4b). These reductions are driven primarily by declines in irrigation water use, reflecting land-use shifts that reduce agricultural production in most LM scenarios (Figure 5a). This reflects the contraction in arable land under LM (Figure 2d,e,g,h), since GCAM's irrigation demand scales with irrigated crop area and crop-specific water requirements. The partial recovery in irrigation withdrawals after 2030 mirrors the corresponding rebound in agricultural production (Figure 3), as continued demand growth presses against the land constraint imposed by LM.

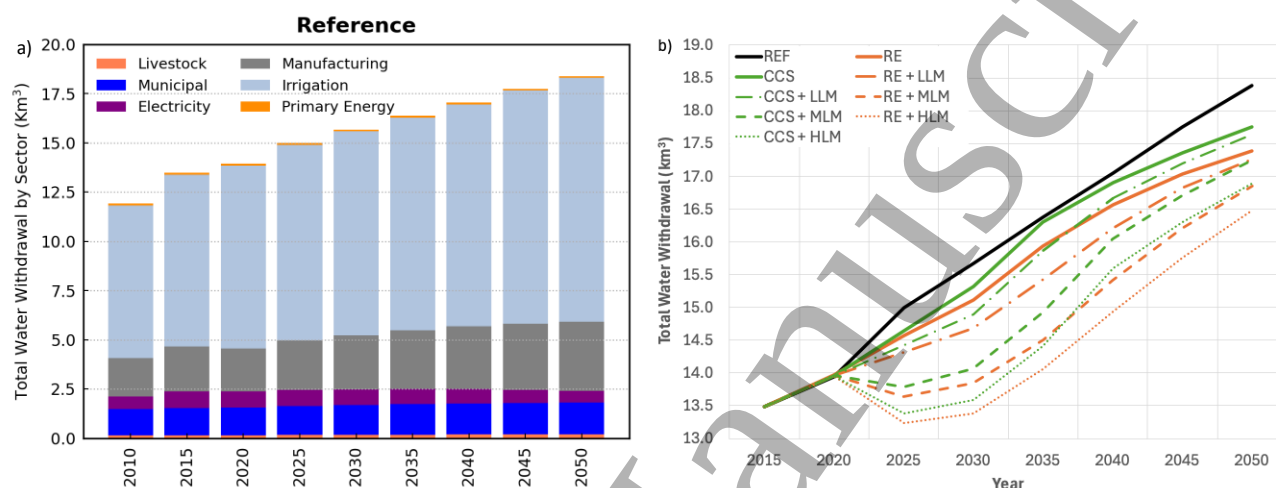


Figure 4. Total water withdrawal (km^3) by sector for the Reference scenario (REF, no climate policy) in panel (a), and total water withdrawal across all carbon-neutrality pathways in panel (b), for Chile 2015–2050. Agricultural irrigation dominates withdrawals across all scenarios, while mitigation pathways reduce total demand relative to REF.

As noted in Figure 1a, Chile's energy portfolio already incorporates renewables in REF to meet increasing electricity demand. Consequently, electricity water withdrawal decreases by 28% in REF (Figure 5b) between 2015 and 2050. GCAM also assumes continued improvement in cooling technologies, with all regions globally transitioning away from once-through flow (OTF) cooling over time [26]. Note, however, that electricity water withdrawal is substantially smaller in scale than irrigation withdrawal (Figure 5).

Electricity-related water withdrawals in the RE scenario are about 72% lower than in REF by 2050, whereas the CCS scenario achieves a smaller reduction of 58% relative to REF (Figure 5b). These reductions arise principally from changes in the composition of electricity generation rather than from a contraction in total electricity output, which remains comparably high across the mitigation scenarios (Figure 1). In the RE pathway, wind and solar replace a substantial share of thermoelectric generation and therefore avoid the cooling-water requirements associated with thermal power plants. In the CCS pathways, withdrawals also decline relative to REF because coal is phased out and the thermal fleet evolves towards lower-emission technologies. However, these reductions are partly offset by continued fossil generation and the

deployment of water-intensive CCS technologies. The comparison then demonstrates that the water implications of decarbonization depend not only on how much electricity is generated, but also on the technologies used to generate it and their associated cooling requirements.

RE + LM scenarios exhibit larger decreases in total and irrigation water withdrawals than their CCS + LM counterparts because they combine low-water electricity generation with the strongest reductions in agricultural land and production (Figures 4b and 5a). By contrast, the availability of CCS reduces the mitigation burden placed on the land sector, resulting in comparatively smaller contractions in agricultural land and irrigation demands. Importantly, the value of this comparison lies not simply in showing that reduced agricultural production lowers irrigation demand, but also, and critically, in quantifying the relative influence of alternative energy and land mitigation pathways on national water withdrawals. It should also be noted that although renewable deployment produces substantial proportional reductions in electricity-sector withdrawals, these remain small compared with irrigation demands, which continue to dominate total water withdrawals across all scenarios. Consequently, national water-demand outcomes appear considerably more influenced by land-

allocation responses than to the choice between renewable- and CCS-based electricity pathways.

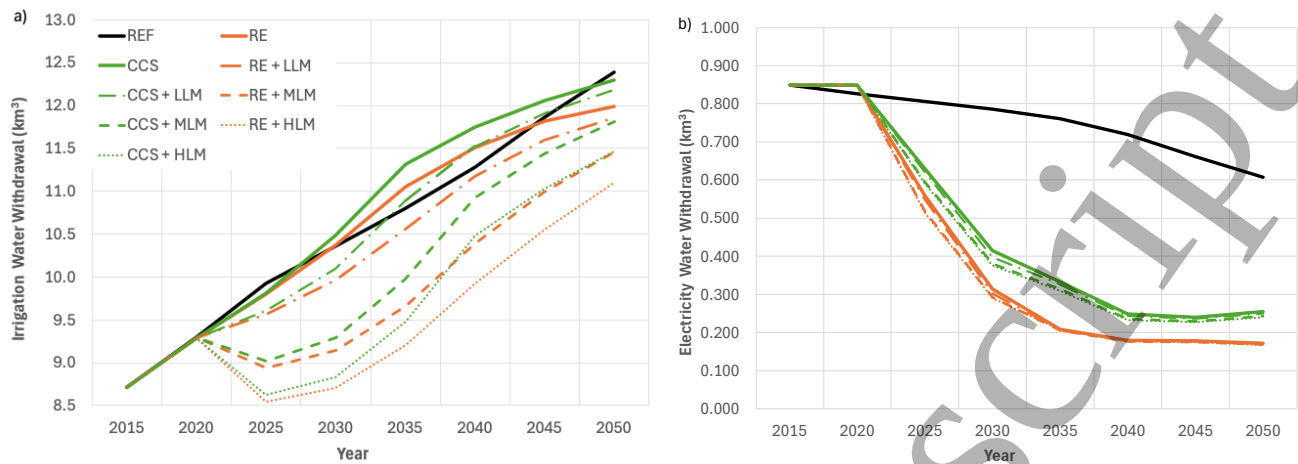


Figure 5. Irrigation (a) and electricity (b) water withdrawal (km^3) across all scenarios in Chile. Scenarios without land mitigation (Reference [REF], Renewables [RE], Carbon Capture and Storage [CCS]) show steady growth in irrigation demand, while Moderate and High LM scenarios exhibit temporary declines reflecting land reallocation under terrestrial carbon pricing. RE scenarios achieve greater reductions in electricity water withdrawal than CCS scenarios by the mid-2020s, reflecting the higher water demands of CCS-associated generation technologies, while both show marked declines relative to REF.

4. Conclusion

Three core insights emerge from simulating alternative carbon-neutrality pathways for Chile within the water-energy-land nexus.

First, renewables-dominated energy-sector pathways substantially reduce electricity-sector water withdrawals, while CCS-enabled pathways achieve smaller reductions due to the water demands of CCS-associated generation technologies. Across scenarios, phasing out coal and shifting toward wind and solar consistently lowers cooling-related water demand, resulting in consistently lower electricity-sector water demand across the scenarios examined. This demonstrates that growth in electricity generation can be partially decoupled from freshwater withdrawals through power-sector decarbonization, even under conditions of increasing electricity demand. These findings align with and extend prior global assessments: while CCS-enabled pathways do reduce electricity water withdrawals relative to REF through coal phase-out. However, the deployment of water-intensive CCS-associated technologies partially offsets these gains, consistent with existing projections of increased water demand under CCS policies [25].

This finding is particularly relevant beyond Chile, as many water-stressed regions face similar trade-offs between climate mitigation and water scarcity in their electricity systems. Should the generation mix remain partially thermal in the longer term, coastal siting with seawater or brackish-water cooling could substantially reduce freshwater competition,

although associated environmental and siting implications would require careful assessment.

Second, total water withdrawals remain dominated by irrigation across all carbon-neutrality pathways, even as power-sector water use declines. Land-sector mitigation moderates water withdrawals by reducing agricultural land, production and associated irrigation demand, but this apparent water-demand co-benefit is inseparable from potentially important economic and food-system trade-offs. While GCAM represents agricultural markets and trade at an aggregate level, it does not capture institutional feasibility, transition costs, distributional food-security outcomes or impacts on rural livelihoods in sufficient detail to assess how such changes could be implemented in practice. The reduction in agricultural production should therefore be interpreted as an unintended consequence of land competition induced by the land-mitigation scenarios, rather than as a desirable or readily implementable pathway. Improvements in agricultural productivity and water-use efficiency could help mitigate this trade-off, while changes in trade patterns may provide only a partial adjustment and could increase exposure to international food-price volatility. The land-mitigation scenarios should therefore be interpreted as exploratory policy pathways that illustrate the magnitude of potential trade-offs instead of direct policy prescriptions.

Third, the combination of energy and land policies reshapes trade-offs across the water-energy-land nexus, rather than delivering uniformly synergistic outcomes. These cross-sector interactions demonstrate that climate policies implemented in isolation risk shifting pressures rather than

resolving them, which is particularly important in water-stressed contexts such as Chile.

Taken together, these results highlight that Chile's dual decarbonization strategy, combining energy and land mitigation, can substantially alter water demand patterns across sectors, producing both benefits and new tensions. The strong reduction in electricity-sector water withdrawals through renewables expansion offers a clear opportunity for water-stressed countries. By contrast, land-based mitigation introduces more complex trade-offs requiring careful policy coordination and depends on the continued viability of terrestrial carbon sinks that are themselves vulnerable to wildfire and climate change. These findings underscore the importance of integrated energy and land policy design in achieving climate goals without adding to water-demand pressures in an already water-stressed context.

A limitation of this analysis is that land mitigation is represented through economic incentives within the GCAM framework. In practice, implementation would also be shaped by institutional capacity, land tenure, governance and social acceptance, which are not explicitly represented in the model. The nine-scenario matrix provides a structured assessment of whether the central findings persist across contrasting energy pathways and levels of land-mitigation stringency. Consistent with the exploratory intent of this study, the results should be read as conditional on the model's baseline parameterization. No formal sensitivity analysis was undertaken, and the results are therefore not shown to be robust to uncertainty in water-use intensities, crop yields, land-allocation responses, or socioeconomic drivers. Formal sensitivity or uncertainty analysis remains an important direction for future work [55–57].

This analysis is also subject to uncertainty in water-use intensities for energy and land-based activities, crop-yield responses, land-sector sequestration potentials, socioeconomic drivers and technology performance. The projected water co-benefits of land mitigation further assume that terrestrial carbon sinks remain effective over time. Increasing wildfire risk could reduce this sequestration potential, diminish the contribution of land mitigation to carbon neutrality and shift a greater share of the mitigation burden towards other sectors, thereby altering the associated water implications. Additional uncertainty arises from the structural characteristics of integrated assessment models, including sectoral aggregation, parameter calibration and the representation of cross-sectoral interactions. The results should therefore be interpreted as scenario-based insights into relative cross-sectoral trade-offs and not as precise development projections. A further limitation is that the analysis does not explicitly model future climate-induced changes in hydrological supply or irrigation requirements. This omission likely leads to an understatement of future water stress. As noted previously, central Chile has already experienced precipitation deficits of approximately 25–

45% during the ongoing mega-drought [11,12], while the LM scenarios reduce national total withdrawals by approximately 4–10% relative to REF by 2050. While these quantities are not directly comparable, they indicate that lower water demand under mitigation does not necessarily imply lower overall water stress under continued drying. Thus, the demand-side benefits identified here may moderately alleviate the pressure on Chile's water system, but they are unlikely to offset the broader water-stress challenge posed by continued drying. This distinction is important because energy- and land-sector mitigation policies, as explored here, can help reduce pressure on the water system, but on their own may not be sufficient to alleviate Chile's underlying structural water scarcity.

Future work should extend the current national-scale GCAM-Chile framework by developing and calibrating a sub-national version integrated with basin-level hydrological projections. Such a framework would better capture regional heterogeneity in renewable-energy potential, land productivity, water demand and water availability across Chile's river basins, providing a more spatially explicit assessment of climate mitigation pathways. It would also enable direct evaluation of whether the water-demand co-benefits identified in the present scenarios remain robust when combined with projected climate-driven reductions in water availability, particularly in Chile's already water-stressed basins. Further methodological improvements could include a more detailed representation of land-allocation constraints, crop-specific irrigation requirements and crop-yield responses to strengthen assessments of land-mitigation impacts on agricultural production and water use. Improved observational water-use data, including re-checking imputed database entries against new national reporting, would also strengthen calibration and validation of GCAM-Chile and similar models. Finally, extending this framework to other water-scarce regions would enable comparative analyses of how different development pathways shape water-energy-land trade-offs under climate mitigation strategies.

Acknowledgements

Felipe Feijoo was funded by the Agencia Nacional de Investigación y Desarrollo (ANID) of Chile by the grant FONDECYT regular 1262572. Pacific Northwest National Laboratory is operated by the Battelle Memorial Institute for DOE under Contract DE-AC05-76RL01830. Francisco Flores has been funded by the European Union (NextGenerationEU) under the National Recovery and Resilience Plan 2021–2026 (NRRP), through the UNIZAG FSB institutional project "Energy Transition of Hard-to-Abate Sectors", approved by the Ministry of Science, Education and Youth of the Republic of Croatia (component C3.2, source 581). The findings and conclusions in this publication are those of the authors alone and

should not be construed to represent any official US Government determination or policy, or the views of any of the institutions associated with the authors of this study. The publisher, by accepting the article for publication, acknowledges that the US Government retains a non-exclusive, paid-up, irrevocable, worldwide license to publish or reproduce the published form of this work, or allow others to do so, for US Government purposes.

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Accepted Manuscript

Glossary

Concept	Acronym
Carbon dioxide	CO ₂
Greenhouse Gas	GHG
Renewable Energy	RE
Carbon Neutrality	CN
Carbon Capture & Storage	CCS
Direct Air Capture	DAC
Nationally Determined Contributions	NDC
Chile's Long-Term Energy Plan (Planificación Energética de Largo Plazo)	PELP
Integrated Assessment Model	IAM
Global Change Analysis Model	GCAM
Water-energy-land	WEL
Land-sector Mitigation	LM
Land use, land use change and forestry	LULUCF
Limited Land Mitigation scenarios	LLM
Moderate Land Mitigation scenarios	MLM
High Land Mitigation scenarios	HLM